


$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$$

$$E[\hat{\beta}_1 | X's] = E\left[\frac{s_{xy}}{s_{xx}} \mid X's\right]$$

$$= \frac{1}{s_{xx}} E[s_{xy} | X's]$$

$$= \frac{1}{s_{xx}} E\left[\sum_i (x_i - \bar{x})(y_i - \bar{y}) \mid X's\right]$$

$$= \frac{1}{s_{xx}} E\left[\sum_i (x_i - \bar{x}) y_i \mid X's\right]$$

$$= \frac{1}{s_{xx}} \sum_i (x_i - \bar{x}) E[y_i | X's]$$

$$= \frac{1}{s_{xx}} \sum_i (x_i - \bar{x}) E[y_i | x_i]$$

$$E[\hat{\beta}_1 | X's] \neq \beta_1 \approx -0.0909 \text{ (from lecture 2)}$$

In the lecture

$$E[Y|X] = \beta_0 + \beta_1 X$$

$$\Rightarrow E[\hat{\beta}_1 | X's] = \beta_1$$

In this problem, the above is NOT TRUE!

$$E[Y|X] = \frac{3x+2}{6x+2}$$

$$\begin{aligned} \sum_i (x_i - \bar{x})(y_i - \bar{y}) &= \sum_i (x_i - \bar{x}) y_i \\ &= \sum_i (y_i - \bar{y}) x_i \end{aligned}$$

$$\sum_i (x_i - \bar{x}) = 0$$

$$f(y_i | x_1, \dots, x_n) = \frac{f(y_i, x_1, \dots, x_n)}{f(x_1, \dots, x_n)}$$

$$= \frac{f(y_i, x_1) \times f(x_2) \times \dots \times f(x_n)}{f(x_1) \times \dots \times f(x_n)}$$

$$\int f(y_1, \dots, y_n | x_1, \dots, x_n) d\mathbf{y}_{-i} = \int f(y_1 | x_1) \times \dots \times f(y_n | x_n) d\mathbf{y}_{-i}$$

$$\nearrow \mathbf{y}_{-i} = (y_1, \dots, y_{i-1}, y_{i+1}, \dots, y_n)$$

everything except y_i

$$\frac{(n-2)\hat{\sigma}^2}{\sigma^2} | X's \sim \chi^2(n-2)$$

$$SE(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

$$\widehat{\text{Var}(\hat{\beta}_1 | X's)} = \frac{\hat{\sigma}^2}{S_{xx}}$$

↑ estimate

$$\hat{\beta}_1 | X's \text{ is Normal } (\beta_1, \text{Var}(\hat{\beta}_1 | X's))$$

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 | X's)}} \sim N(0, 1)$$

$$\text{Recall } \text{Var}(\hat{\beta}_1 | X's) = \frac{\sigma^2}{S_{xx}}$$

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / S_{xx}}} \sim N(0, 1)$$

$$\frac{\hat{\beta}_1 - \beta_1}{\frac{\sigma^2}{\sqrt{S_{xx}}}} \sim N(0, 1)$$

$$\text{Cov}(\bar{Y}, \hat{\beta}_1 | X's) = \text{Cov}(\hat{\beta}_0 + \hat{\beta}_1 \bar{X}, \hat{\beta}_1 | X's)$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \Rightarrow \bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$$
